

Industrial Control Systems, ICS700 6 credits and ADB600 7,5 credits

This course PM applies to the course instance: VT-2, 2026, study period 3)

The purpose of the course PM is to provide you, as a student, with the information you need to plan your studies. The information in the course syllabus always takes precedence over the course PM.

For course objectives, course content, and entry requirements, see the current course syllabus. For literature, see the course literature. Both documents are available on Canvas in the Course Information module. The current schedule is available at schema.hv.se (Kronox), search for the course code.

Faculty contact information

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Course structure and pedagogical approach

The course is based on theory lectures, followed by assignments in which students independently design, simulate, and program a PLC in TwinCat, or, on some occasions, write a page to hand in on Canvas. Each week consists of theory lectures and one assignment linked to the theory.

The theory lectures are streamed in real-time and recorded for distance students.

The examination is based on the assignments, a PLC exam, and a written examination.

Students taking ADB600 are working on a big data project at the end of the study period.

Week 4 Start week

Theory: Lecture 1, Theory Intro to Automation Technique and PLC

Lecture 2, Boolean Logic LD, FBD, ST

Preparation: Install TwinCat on your Windows computer and learn how to use it

Week 5

Assignment: Combined Logic (Elevator)

Extra lab assistance occasion on Zoom and in the Computer room

Week 6

Theory: Lecture 3, Sequence programming, SFC, ST

Assignment: Sequence programming (Washing Machine)

Week 7

Theory: Lecture 4, Integer, Float, variables, operations, and reusable function blocks

Assignment: Control of tank level

Week 8

Theory: Lecture 5, Manual/Auto and Programming Structures

Lecture 6, Machine safety and Risk reduction

Assignment: Structured programming (Drilling Machine). Hand the PLC solution and a **video** explaining the drilling machine functionality from a safety perspective. You **must be visible** on the video at times when you explain the functionality.

Week 9

Theory: Lecture 7, Communication and handshake

Assignment: Supervisory control and virtual process

Week 10

Theory: Lecture 8, Overview of field buses

Practical Exam: A PLC task on a TwinCAT simulation

Week 11

Theory: Lecture 9, IoT and Implementations

Assignment: Hand in a written page of an IoT protocol

Week 12

Theory: Lecture 10, Creation of a TwinCat project from scratch and HMI

Assignment: HMI (AGV)

Week 13

Examination: Written examination as a quiz in Canvas

Week 16

Practical Re-Exam: A PLC task on a TwinCAT simulation

Re-Examination: Written examination as a quiz in Canvas

Complete course grading

- **Approved assignments (U/G)**
- **The video on the drilling machine and safety is graded (U/G/VG)**
- **Practical PLC exam (U/G/VG)**
- **The written examination (quiz) is graded (U/G/MVG)**

The following table weights the final grade for the whole course:

	E	D	D	D	C	C	C	B	B	B	A
Assignments	G	G	G	G	G	G	G	G	G	G	G
Video on drilling m.	G	VG	G	G	G	VG	G	VG	G	VG	VG
Practical PLC exam	G	G	VG	G	VG	G	G	VG	VG	G	VG
Written exam, quiz	G	G	G	VG	VG	VG	MVG	VG	MVG	MVG	MVG

Use of generative AI in the course

Generative AI, such as ChatGPT, Copilot, or others, can be used to correct text. Generative AI is **not allowed** to be used to find answers in the Quiz and the Practical PLC exam.

Technical prerequisites and equipment

At University West, there are a number of computer rooms with software required by courses, as well as course-specific software such as TwinCAT. It is possible to download software (such as MS Office and TwinCAT) to a private computer as a student, see hv.se. Those are Windows applications. If you, as a student, choose another solution, e.g., a mobile device, iPad, or similar, you are responsible for ensuring the necessary functionality. Examples of standard submission formats are DOCX, PPTX, XLSX, and PDF.

Course evaluation

The formal process the university uses to ensure that students' views, linked to course evaluations, are used and contribute to course development assumes that students complete the course evaluation. Student representatives are part of the program council, department committee, and research and education committee. Student engagement is essential. Contact the student union if you want to be a student representative.

It is important as a student to do course evaluation. The students' course evaluation answers are compiled and commented on by the examiner and course coordinator in a course evaluation response that is published on the course's Canvas page. The course evaluation response is also communicated and discussed in the program council. The Program Council compiles an annual program report based on all course evaluation responses. This program report is presented and discussed at the department committee meeting. The department committee compiles all program reports into a department-wide

quality report, which is discussed by the research and education committee, communicated to department management, and reported back to the program councils. The Research and Education Board compiles the quality reports of all departments in a university-wide quality report which is communicated to the university's top management and is reported back to all department boards.

Information on misleading of examination (cheating)

Misleading of examination, so called cheating, can, with the support of the Higher Education Ordinance (1993: 100), lead to a written warning or suspension from studies. Therefore, it is important that you are aware of what is not allowed.

Cheating is about trying to make an examination performance appear bigger, better or in some other way of higher quality than would otherwise have been the case. This can, for example, be about plagiarism, the use of unauthorized aids, illicit cooperation, etc. See more information at <https://www.hv.se/en/student/studies/examination/cheating-and-plagiarism/>

If you use text that someone else has written (from literature, other student work, internet, etc.), you must clearly state the source. The same goes for direct translations. Quotes must be marked with quotation marks. The examiner must be able to clearly see what is your own text and what is written by others. If a source reference or quotation mark is missing, it may be counted as plagiarism, which may lead to you being suspended from your studies.

If you reuse a text that you have previously been examined for, it may be considered self-plagiarism. Therefore, be sure to clearly state whether your text has already been used by you in another examination.

To detect plagiarism, Högskolan Väst uses "Ouriginal", a software for plagiarism control. Ouriginal's source material is extensive and consists of both published and unpublished texts, such as previous student's work.

Teachers who suspect misleading of examination are required to report this to the Vice-Chancellor, who decides if it should be referred to the Disciplinary Board at University West. <https://www.hv.se/en/meet-university-west/organisation/boards-councils-and-committees/disciplinary-board/>

Support for students with disabilities

Contact the coordinators for pedagogical support at HV before the start of the course if you, as a student, are in need of any support. <https://www.hv.se/en/student/support-and-service/students-with-disabilities/>

In order to receive support, you must have a certificate stating that your disability is permanent or long-term and you as a student need to apply to be able to receive support.